



EG1113 - C Programming for Engineering Lab

Report #3

Pulse Width Modulation to the Screen and an LED

Date of the experiment: September 4, 2024

Due date of report: September 6, 2021

Name: Nicholas Galus

1. Core concepts learned

Summarize the core concepts learned and practiced in this laboratory.

- Setup remote desktop to utilize the assigned Raspberry Pi 400
- Established network file sharing between the Raspberry Pi 400 and laptop utilizing Caja
- Created a hello world program in the C programming language

2. List of equipment and software used

- Raspberry PI 400
- Notepad
- Lenovo Laptop
- Caja File Network Share
- Visual Studio Code
- Protoboard with buttons/lights

3. Relevant web links and other documents:

- Utilized starter code that was provided with the lab to test button and lights

4. Exercise: Pulse

4.1 Summary

- Learning about Pulse Width Modulation and answer questions about it

4.2 Procedures

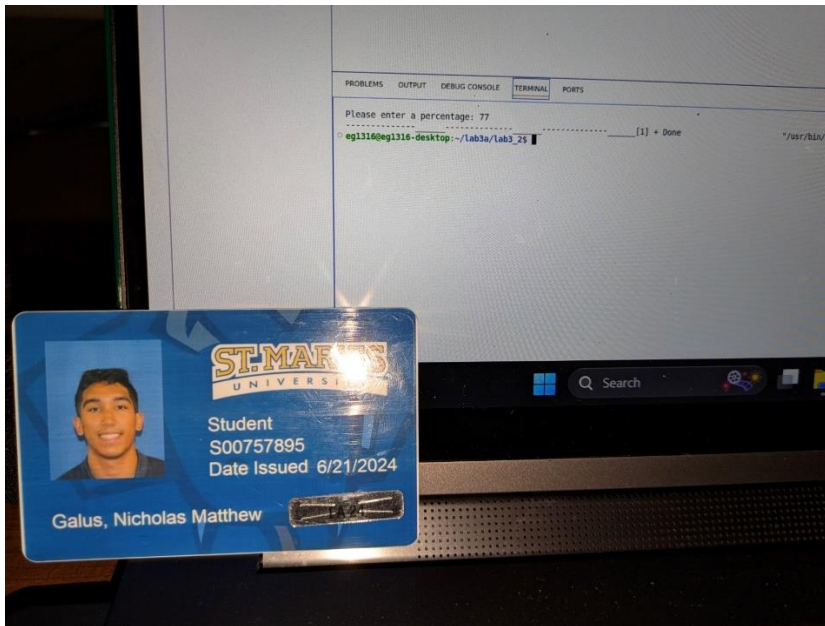
1. How could the “percentage” the user provides in the program you wrote in exercise 3 (below) be related to the speed of something like Joe’s car?
The percentage controls the average speed of the vehicle. By holding the throttle down for a certain percentage of a period, you can have varying average speeds.
2. If we call the height of the “-” character 100 and the height of the “_” character 0 for the output of your program, what is the average height of all of the “_” and “-” characters printed when the user enters 73?
The average height would be 70% as displayed as the following:
3. How could you use this average height concept to control the speed of a motor?
By using the average height concept, you can control the speed of a motor by sending pulses of power to it depending on the percentage of the period, which will control the speed of the motor.

4. How could you use it to control the position (how open it is) of a valve?
You can use PWD to control the position of a valve by opening it based on the average height of the period.

5. Exercise: PWD Output to Screen

5.1 Summary

- Create a program to display Pulse Width Modulation
- Lab Verification:



5.2 Procedures

1. Create a program to display Pulse Width Modulation with the use of “—” and “__” based on an input

```
1  #include <stdio.h>
2
3  int main(void)
4  {
5      int percentage;
6      int i, up_count, down_count, cycles;
7
8      printf("Please enter a percentage: ");
9      scanf("%d", &percentage);
10
11     if((percentage < 0) || (percentage > 100))
12         printf("Sorry, that number is not valid! Bye.");
13
14     up_count = percentage / 10;
15     down_count = 10 - percentage / 10;
16
17     for(cycles = 0; cycles < 3; cycles++)
18     {
19         for(i = 0; i < up_count; i++)
20             printf("--");
21
22         for(i = 0; i < down_count; i++)
23             printf("__");
24     }
25 }
```

2. Test the every possible input:

```

16   for(int j = 0; j <= 100; j += 10)
17   {
18       percentage = j;
19
20       up_count = percentage / 10;
21       down_count = 10 - percentage / 10;
22
23       for(cycles = 0; cycles < 3; cycles++)
24       {
25           for(i = 0; i < up_count; i++)
26               printf("--");
27
28           for(i = 0; i < down_count; i++)
29               printf("__");
30       }
31
32       printf("\n");
33   }

```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

[illegible]

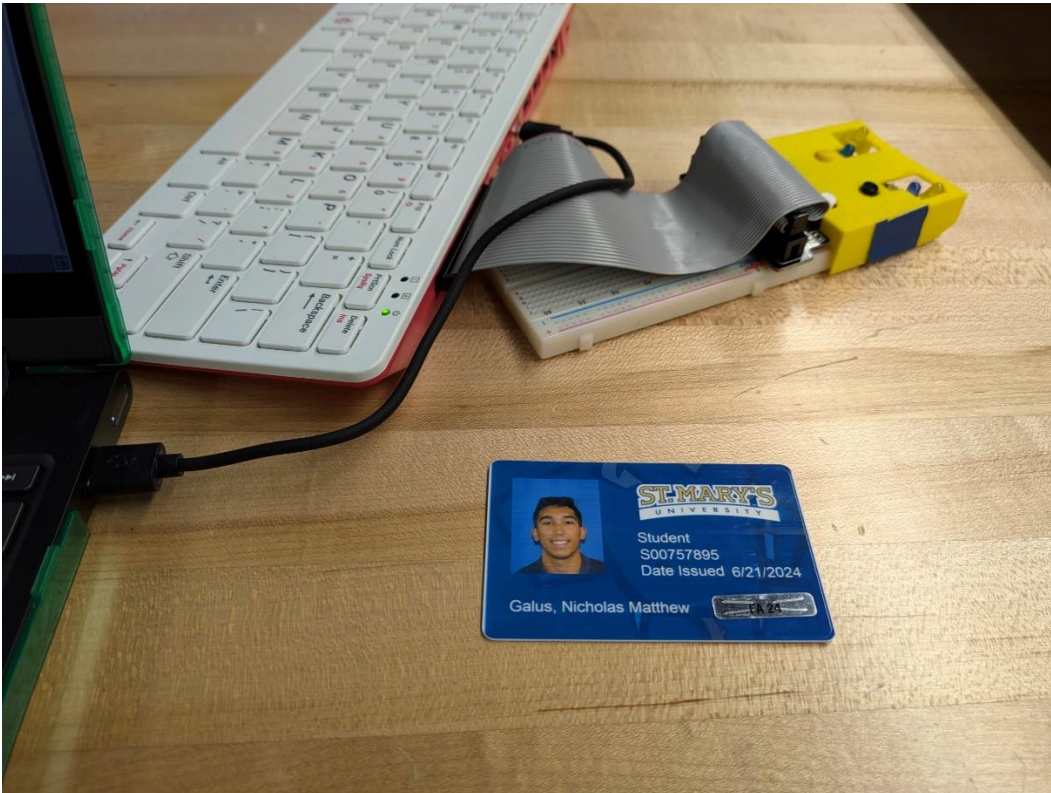
6. Exercise: Protoboard Setup/Button Tester

6.1 Summary

- Connecting the protoboard to

6.2 Procedures

1. Make sure pi is powered off and disconnected from power (USB-C cable)
2. Plug in the ribbon cable



3. Power PI on and compile/run the button tester program

```

eg1316@eg1316-desktop:~/lab3/lab3_3$ gcc -o button_tester button_tester.c -lpigpiod_if2 -lraylib -lm
eg1316@eg1316-desktop:~/lab3/lab3_3$ ./button_tester 127.0.0.1
connecting local buttons to 127.0.0.1
connecting to lights (or car at) 127.0.0.1

***** PIGPIO connection failed! *****
***** Did you execute 'sudo pigpiod'? *****

eg1316@eg1316-desktop:~/lab3/lab3_3$ gcc -o button_tester button_tester.c -lpigpiod_if2 -lraylib -lm
eg1316@eg1316-desktop:~/lab3/lab3_3$ ./button_tester 127.0.0.1
connecting local buttons to 127.0.0.1
connecting to lights (or car at) 127.0.0.1

***** PIGPIO connection failed! *****
***** Did you execute 'sudo pigpiod'? *****

eg1316@eg1316-desktop:~/lab3/lab3_3$ sudo pigpiod
[sudo] password for eg1316:
eg1316@eg1316-desktop:~/lab3/lab3_3$ ./button_tester 127.0.0.1
connecting local buttons to 127.0.0.1
connecting to lights (or car at) 127.0.0.1

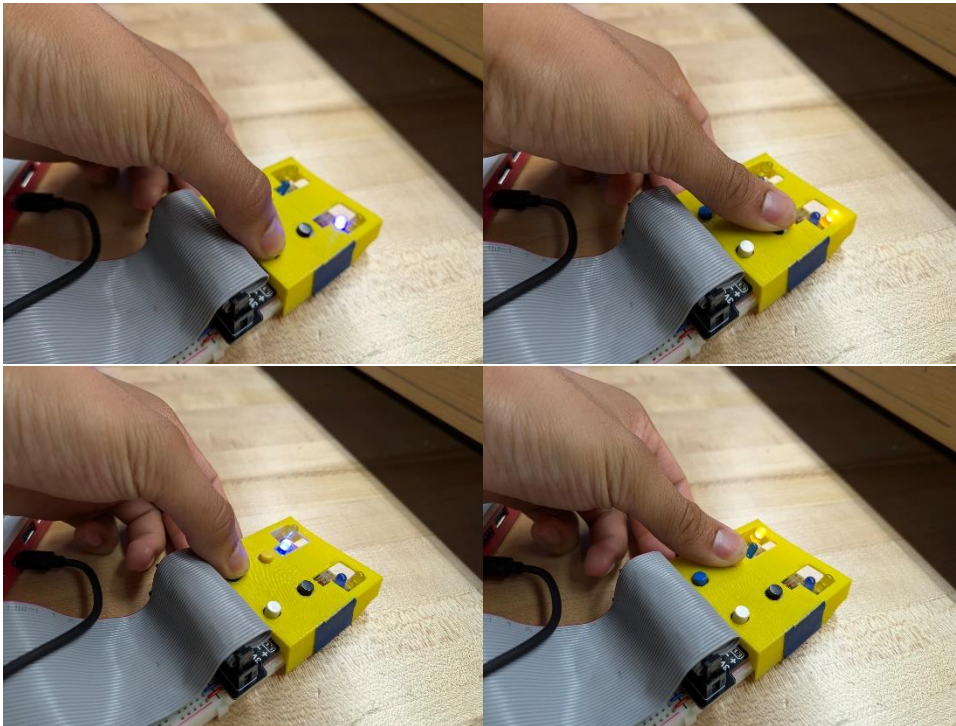
Checking for Wireless connection information
vlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.102.22.222 netmask 255.255.0.0 broadcast 10.102.255.255

Successfully connected to the buttons and a car (or car simulator)...here we go
]

```



4. Test each button to ensure full functionality



7. Exercise: Blinking light

7.1 Summary

- Modify the exercise 4 starter code to make the LED flash 60 times a second
- Lab verification:



7.2 Procedures

1. Modify the starter code to request an inputted percentage between 0 and 100

```

27     printf("Please enter a percentage: ");
28     scanf("%d", &percentage);
29
30
31     if((percentage < 0) || (percentage > 100))
32         printf("Sorry, that number is not valid! Bye.");
33
34

```

2. Have the light flash on and off based on the inputted percentage

```

35     time_on_portion_of_period_in_microseconds = (period_total_in_microseconds * percentage) / 100;
36     time_off_portion_of_period_in_microseconds = (period_total_in_microseconds - time_on_portion_of_period_in_microseconds);
37
38     for(;;)
39     {
40         gpioWrite(led_io_pin, led_on_pin_value);
41         usleep(time_on_portion_of_period_in_microseconds);
42         gpioWrite(led_io_pin, led_off_pin_value);
43         usleep(time_off_portion_of_period_in_microseconds);

```

3. Modify the code, so that the light flashes 60 times in 1 second based on the inputted percentage

```

40         gpioWrite(led_io_pin, led_on_pin_value);
41         usleep((time on portion of period in microseconds/60));
42         gpioWrite(led_io_pin, led_off_pin_value);
43         usleep(time off portion of period in microseconds/60);

```

Problems experienced and areas of concern

- Was entering VS Code through the “code ‘name of program’” instead of “code .”
 - This causes the main directory to be switch to the program directory instead of the folder with all of the necessary information

8. Course goals

Rank the level (0=none to 5=a large amount) that this lab covered the each of the course goals by updating the appropriate box in the table below.

[illegible]